# RETAIL BUSINESS PERFORMANCE & PROFITABILITY ANALYSIS

## Project Report

### Submitted By

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# 1. Abstract

In today's competitive business environment, organizations generate a large amount of transactional data through sales, customers, products, and different geographical regions. Analyzing this data is important for understanding business performance and identifying areas that require improvement.

The **Retail Business Performance & Profitability Analysis** project focuses on analyzing retail transactional data to understand sales performance, profitability, customer segments, product performance, regional performance, and trends over time. The project follows an end-to-end data analytics workflow starting from data loading and cleaning to analysis and visualization.

Python was used for data cleaning, validation, exploratory data analysis, and calculation of important business metrics. MySQL was used to perform SQL-based analysis and aggregate business data. Tableau was used to create multiple interactive visualizations and a final dashboard that provides a consolidated view of the retail business.

The analysis helps identify high-performing and low-performing categories, regions, customer segments, and products. It also helps identify products or sub-categories that may negatively affect profitability. The final outcome of the project is an interactive business dashboard and a set of analytical insights that can support data-driven decision-making.

**Keywords:** Data Analytics, Retail Analysis, Python, SQL, MySQL, Tableau, Sales Analysis, Profitability Analysis, Business Intelligence.

# 2. Introduction

Retail businesses generate large amounts of data from customer orders, products, sales transactions, discounts, and profits. Raw data by itself does not provide useful business information unless it is properly cleaned, analyzed, and visualized.

Business organizations need answers to questions such as:

- Which product categories generate the highest sales?
- Which categories generate the highest profit?
- Which regions perform better?

- Which customer segments contribute the most?
- Which products generate high sales?
- Which products are loss-making?
- How does sales performance change over time?
- Does high sales always result in high profit?

This project addresses such questions by performing a detailed analysis of retail transactional data.

The project uses three important technologies:

- **Python** for data processing and exploratory data analysis.
- **MySQL** for structured SQL-based analysis.
- **Tableau** for creating visualizations and a final business dashboard.

The combination of these tools demonstrates a complete data analytics workflow.

# 3. Project Objectives

The major objectives of the project are:

1. To analyze the overall performance of the retail business.
2. To calculate important business Key Performance Indicators (KPIs).
3. To analyze sales and profit by product category.
4. To analyze performance at the sub-category level.
5. To identify profitable and loss-making products.
6. To analyze sales and profit across different regions.
7. To analyze the performance of different customer segments.
8. To study sales trends over time.
9. To perform SQL-based business analysis using MySQL.
10. To create interactive visualizations using Tableau.
11. To develop a final dashboard containing important business insights.
12. To provide recommendations that can support business decision-making.

# 4. Tools and Technologies Used

| Tool / Technology | Purpose |
|---|---|
| Python | Data processing and analysis |
| Pandas | Data manipulation and aggregation |
| Matplotlib | Data visualization |
| Google Colab | Python notebook execution |
| MySQL | Database management |
| MySQL Workbench | SQL query execution |

| Tool / Technology | Purpose |
|---|---|
| SQL | Business data analysis |
| Tableau | Data visualization and dashboard creation |
| CSV | Dataset storage |

# 5. Dataset Description

The project uses retail transactional data containing information related to orders, customers, products, sales, discounts, and profit.

The major attributes in the dataset include:

- Order ID
- Order Date
- Ship Date
- Customer ID
- Customer Name
- Segment
- Country
- City
- State
- Region
- Product ID
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit

These attributes make it possible to analyze retail performance from different perspectives.

For example:

- **Category and Sub-Category** are used for product performance analysis.
- **Region** is used for geographical analysis.
- **Segment** is used for customer analysis.
- **Sales and Profit** are used to measure business performance.
- **Order Date** is used for time-based analysis.
- **Discount** can be analyzed to understand its relationship with profitability.

# 6. Project Methodology

The project follows the following data analytics workflow:

Retail Dataset

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Data Loading

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Data Understanding

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Data Cleaning and Validation

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Feature Engineering

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Exploratory Data Analysis

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Business KPI Analysis

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Python Analysis

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SQL Analysis

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Tableau Visualization

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Final Dashboard

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Business Insights and Recommendations

# 7. Data Loading and Understanding

The dataset was initially loaded into Python using the Pandas library.

The following operations were performed:

- Viewing the first few records using head().
- Checking the dataset dimensions.
- Viewing column names.
- Checking data types.
- Generating dataset information.
- Checking for missing values.

Example operations include:

```
df.head()
df.shape
df.columns
df.info()
df.isnull().sum()
```

This initial step is important because it helps understand the structure and quality of the dataset before analysis.

# 8. Data Cleaning and Validation

Data cleaning is an important stage in any data analytics project because incorrect or invalid data can produce inaccurate results.

The following validation checks were performed:

### 8.1 Missing Value Check

The dataset was checked for missing values to identify incomplete records.

### 8.2 Duplicate Record Check

Duplicate records were checked to prevent incorrect aggregation of sales and profit.

### 8.3 Date Conversion

The order date and shipping date were converted into datetime format to support time-based analysis.

### 8.4 Sales Validation

Sales values were checked for invalid or unexpected values.

### 8.5 Quantity Validation

Quantity values were checked to ensure that the values were valid for analysis.

### 8.6 Discount Validation

Discount values were examined before performing profitability analysis.

### 8.7 Shipping Validation

Shipping days were calculated using:

Shipping Days=Ship Date−Order Date\text{Shipping Days} = \text{Ship Date} - \text{Order Date}

The data was also checked for invalid negative shipping durations.

# 9. Feature Engineering

Additional features were created from the existing data to support better analysis.

The created features include:

- Order Year
- Order Month
- Quarter
- Seasonal information
- Shipping Days

These features help perform:

- Monthly analysis
- Yearly analysis
- Seasonal analysis
- Shipping performance analysis

Feature engineering improves the ability to identify hidden patterns in the data.

# 10. Business KPI Analysis

Key Performance Indicators were calculated to obtain an overall summary of business performance.

The major KPIs include:

### 10.1 Total Sales

Total revenue generated by the retail business.

### 10.2 Total Profit

Total profit generated after considering product profitability.

### 10.3 Total Quantity

The total number of products sold.

### 10.4 Total Orders

The number of unique customer orders.

### 10.5 Total Customers

The number of unique customers.

### 10.6 Profit Margin

Profit margin measures the percentage of sales converted into profit.

The formula used is:

$$\text{Profit Margin (\%)} = \frac{\text{Total Profit}}{\text{Total Sales}} \times 100$$

Profit margin is important because high sales do not always mean high profitability.

# 11. Category Analysis

The retail data was analyzed based on the major product categories:

- Furniture
- Office Supplies
- Technology

The following metrics were calculated:

- Total Sales
- Total Profit

This analysis helps determine which categories contribute most to:

- Business revenue

- Overall profitability
- Business growth

A category may have high sales but lower profit. Therefore, both sales and profit must be analyzed together.

# 12. Sub-Category Analysis

Sub-category analysis provides more detailed information than category-level analysis.

The following metrics were calculated:

- Total Sales
- Total Profit
- Total Quantity
- Number of Orders
- Profit Margin

The purpose of this analysis is to identify:

- Highly profitable sub-categories
- Low-performing sub-categories
- Loss-making sub-categories

Sub-category analysis is useful because problems at the detailed product level may not be visible when only category-level analysis is performed.

# 13. Regional Performance Analysis

The business performance was compared across different regions:

- Central
- East
- South
- West

The following metrics were analyzed:

- Total Sales
- Total Profit
- Total Quantity
- Number of Orders

Regional analysis helps answer questions such as:

- Which region generates the highest sales?
- Which region generates the highest profit?
- Which regions require improvement?

The results can support better geographical planning, inventory allocation, and marketing strategies.

# 14. Customer Segment Analysis

The dataset contains three main customer segments:

- Consumer
- Corporate
- Home Office

The following metrics were analyzed:

- Total Sales
- Total Profit
- Total Quantity
- Number of Orders
- Number of Customers
- Profit Margin

Customer segment analysis helps understand which group contributes more significantly to the business.

This information can support:

- Customer targeting
- Marketing strategies
- Customer relationship management
- Personalized promotional campaigns

# 15. Product Performance Analysis

Product-level analysis was performed to understand individual product performance.

The following analyses were completed:

### 15.1 Top 10 Products by Sales

Products were ranked based on their total sales.

This helps identify products that generate the highest revenue.

### 15.2 Top 10 Products by Profit

Products were ranked according to their total profit.

This identifies products that contribute strongly to business profitability.

### 15.3 Bottom 10 Products by Profit

Products were ranked according to the lowest profit.

Some products may have negative profit, indicating that they may require further investigation.

Possible reasons for poor profitability include:

- High discounts
- Low selling prices
- High product costs
- Poor pricing strategies

# 16. Sales Trend Analysis

Time-based analysis was performed to understand changes in sales performance.

The analysis can help identify:

- Monthly sales trends
- Changes in business performance over time
- High-demand periods
- Seasonal patterns

Understanding sales trends is useful for:

- Inventory planning
- Marketing campaigns
- Seasonal promotions
- Demand forecasting

# 17. SQL Analysis

After completing data cleaning and preparation, the dataset was exported and imported into MySQL.

SQL was used to perform structured business analysis.

The SQL analysis includes:

- Record verification
- Data aggregation
- Overall KPIs
- Category analysis
- Sub-category analysis
- Regional analysis
- Customer segment analysis
- Product analysis
- Monthly trends
- Discount analysis

Important SQL operations include:

```
SELECT
FROM
WHERE
GROUP BY
ORDER BY
LIMIT
COUNT()
SUM()
AVG()
ROUND()
```

Example category profitability query:

```
SELECT
    category,
    ROUND(SUM(sales), 2) AS total_sales,
    ROUND(SUM(profit), 2) AS total_profit,
    SUM(quantity) AS total_quantity,
    ROUND((SUM(profit) / SUM(sales)) * 100, 2)
        AS profit_margin_percent
FROM retail_sales
GROUP BY category
ORDER BY total_profit DESC;
```

SQL provides an efficient way to summarize and analyze large amounts of transactional data.

# 18. Tableau Visualization

Tableau was used to create business visualizations from the retail dataset.

The final dashboard contains **9 worksheets**:

1. **Sales and Profit by Category**
2. **Sales by Category**
3. **Profit by Category**
4. **Sales by Region**
5. **Profit by Region**

6. **Sales Trend Over Time**
7. **Sales by Customer Segment**
8. **Profit by Customer Segment**
9. **Top 10 Products by Sales**

These visualizations provide a clear and consolidated view of retail business performance.

# 19. Final Dashboard

The final Tableau dashboard combines all major analyses into a single interface.

The dashboard allows users to understand:

- Category performance
- Regional performance
- Customer segment performance
- Sales trends
- Product performance

The dashboard makes complex data easier to understand through charts and visual comparisons.

# 20. Key Business Insights

Based on the analysis performed, the project helps identify the following important insights:

### 1. Sales and Profit Should Be Analyzed Together

A product or category may generate high sales but may not generate high profit.

Therefore, businesses should monitor profitability along with revenue.

### 2. Loss-Making Products Require Investigation

Products with negative or low profit should be investigated.

Possible factors include:

- High discounts
- Low selling prices
- High operational costs

### 3. Regional Performance Can Differ

Different geographical regions may show different sales and profit patterns.

Businesses can use this information to improve weaker regions and strengthen successful regions.

### 4. Customer Segments Have Different Business Value

Different customer groups contribute differently to sales and profitability.

Customer segment analysis can help businesses develop targeted marketing strategies.

### 5. Product-Level Analysis Is Important

Some products may generate strong sales and profit, while others may negatively affect profitability.

Businesses should regularly review their product portfolio.

### 6. Time Trends Support Business Planning

Monthly and seasonal trends can help businesses improve:

- Inventory planning
- Marketing campaigns
- Promotions
- Demand planning

# 21. Business Recommendations

Based on the project analysis, the following recommendations can be considered:

### 1. Review Loss-Making Products

Products with negative profit should be reviewed for:

- Pricing
- Discounts
- Cost
- Demand

### 2. Monitor Profit Margin

Businesses should not focus only on sales.

Profit margin should also be monitored regularly.

### 3. Improve Weak Regions

Regions with lower performance should be investigated to identify possible issues related to:

- Customer demand
- Marketing
- Product availability

### 4. Optimize Discount Strategies

High discounts can increase sales but may reduce profitability.

Discount policies should be reviewed based on product performance.

### 5. Focus on High-Value Customer Segments

Customer segments that contribute strongly to sales and profit can be targeted with specialized marketing strategies.

### 6. Improve Inventory Planning

Historical sales trends can help businesses maintain appropriate inventory levels and avoid:

- Overstocking
- Understocking

# 22. Skills Demonstrated

This project demonstrates skills in the following areas:

### Python

- Pandas
- Data Cleaning
- Data Manipulation
- Data Aggregation
- Exploratory Data Analysis
- Data Visualization

### SQL

- Database Management
- SELECT Queries
- Aggregation Functions
- GROUP BY
- ORDER BY
- Business KPI Calculation

### Tableau

- Data Visualization
- Worksheet Creation
- Dashboard Development
- Business Reporting

### Business Analytics

- KPI Analysis
- Profitability Analysis
- Sales Analysis
- Product Analysis
- Regional Analysis
- Customer Analysis

# 23. Conclusion

The **Retail Business Performance & Profitability Analysis** project demonstrates a complete end-to-end data analytics workflow.

The retail transactional dataset was cleaned, validated, analyzed, and visualized using **Python, MySQL, and Tableau**. The project analyzed important aspects of business performance, including:

- Sales
- Profit
- Product categories

- Sub-categories
- Regions
- Customer segments
- Products
- Time trends

Python was used for data cleaning and exploratory analysis, SQL was used for structured business queries, and Tableau was used to create an interactive dashboard.

The project demonstrates how raw business data can be transformed into meaningful insights that support data-driven decision-making. The final dashboard provides a consolidated view of retail performance and can help businesses identify opportunities for improving profitability, product performance, customer strategies, and operational planning.

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# 24. Project Deliverables

The final project includes:

```
Retail-Business-Performance-Analysis/
│
├── Dataset/
│   └── retail_cleaned_for_sql.csv
│
├── Python_Analysis/
│   └── Retail_Business_Performance_Analysis.ipynb
│
├── SQL_Analysis/
│   └── Retail_Business_SQL_Analysis.sql
│
├── Tableau_Dashboard/
│   └── Retail_Business_Performance.twb
│
├── Report/
│   └── Retail_Business_Performance_Report.pdf
│
└── README.md
```

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